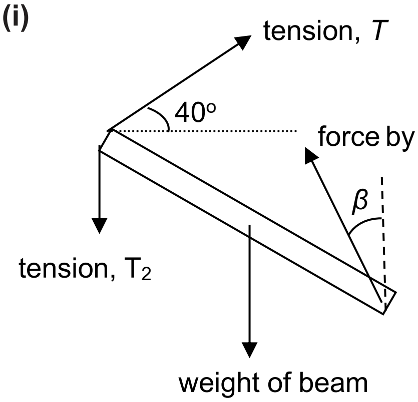
Section A

1 **(a)** Resultant force on it must be zero in any direction. **B1**

Resultant torque on it must be zero about any axis (of rotation). **B1**

MC:	Quite badly done. Missing key words in conditions stated. Marks penalized if there is no mention of about any axis (for torque) or in any direction (for force) in answers.

(b) **(i)**



Any 2 correct pairs of forces **B1**

Other 2 correct pairs of forces **B1**

With correct direction and label

MC:	The question specifically mention to draw in Fig 1.1, but some drew a separate diagram instead. BOD awarded if T_2 is replaced by weight of the 5.0 kg, Force by wall is not a normal force since it is not 90^0. Many students drew the force by the wall wrongly, they either assumed its horizontal, or along the beam.

(ii) Let x be the distance of c.g of beam from the hinge.

Taking moments about hinge,

sum of clockwise moments = sum of anticlockwise moments

$120\ (5 \sin 60^\circ) = 5g\ (5 \sin 70^\circ) + 20g\ (x \sin 70^\circ)$ **C1**

$x = 1.57 =$ 1.6 m **A1**

MC:	Poorly done, because many could not obtain the correct clockwise or anticlockwise moment about the hinge. Some made mistakes when determining the necessary angle needed for calculations.

(iii) Vertical summation of forces:

$F_Y + 120\ (\sin 40^\circ) = 5g + 20g$

$F_Y = 168.1$ N

Horizontal summation of forces:

C1

$$F_X = 120\ (\cos 40^\circ)$$
$$= 91.93\ \text{N}$$

$$F = \sqrt{F_X^{\,2}+F_Y^{\,2}} = \sqrt{91.93^2+168.1^2} = 192\ \text{N}$$ **A1**

$$\tan\beta = \frac{91.93}{168.1}$$

$$\beta = 29^\circ$$ **A1**

Force is at an angle of 41° with the beam **A1**

MC:	Many failed to resolve horizontally and vertically. A few did not calculate the numerical angle of the force by wall, while many did not express the angle correctly with the beam.

2(a)	In uniform circular motion, the **speed of the metal ball is constant**, but its **velocity** is **constantly changing direction**. Since **acceleration is the rate of change of velocity**, the metal ball experiences an acceleration.	**B1** **B1**
	OR In uniform circular motion, the **speed of the metal ball is constant**, but there is a **net non-zero force acting on the metal ball according to Newton's 1st law** because **it is moving in a circular path**. From **Newton's 2nd law for constant mass**, there must be an **acceleration in the same direction as the net force**.	**B1** **B1**

CKW MC	• Most candidates demonstrated a lack of understanding regarding the significance of *uniform circular motion*. Paraphrasing the question, candidates should explain why a metal ball moving at a constant speed in circular motion is still considered to be accelerating. • Some candidates mentioned the presence of a centripetal force without explaining why the centripetal force exists • Some candidates explained why the metal ball experiences a centripetal acceleration and not an acceleration. • Some candidates quoted Newton's 2nd law in general or in mathematical form, without explaining why acceleration cannot be zero. • For the A-level syllabus, because of the general case of variable mass systems, a force can be defined as the rate of change in momentum, and it is the change in momentum that leads to the emergence of a force.	
2(b)(i)		